Student Course Management System

Solo Project & Technical Implementation Review

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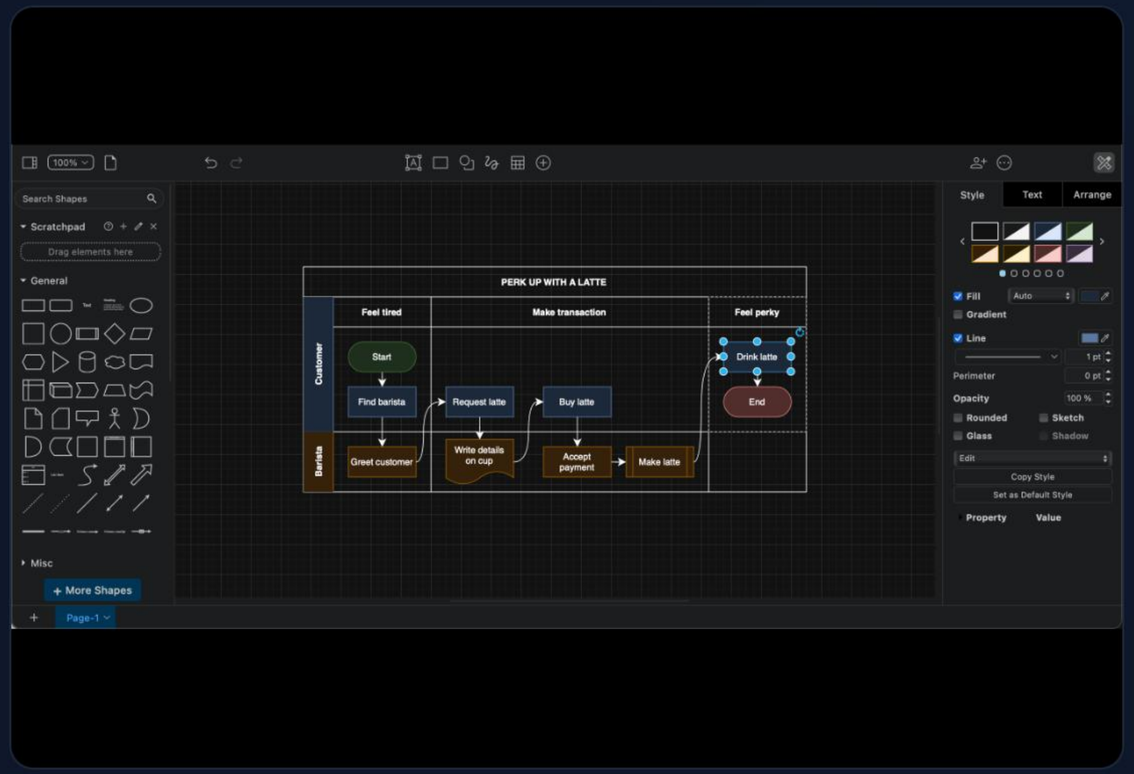
Foundations of Software Development

Project Overview

A highly structured, command-line C++ application leveraging strict Object-Oriented Programming and Standard Template Library containers for efficient academic record management.

Class Hierarchy (OOP)

- **Encapsulation & Inheritance:** The architecture strictly separates core identity from academic metrics.
- **Base Class (Person):** Acts as the foundation, securely maintaining the protected `name` string variable.
- **Derived Class (Student):** Inherits the name attribute and extends the blueprint to include `studentId`, `gradePointAverage`, and an isolated set of enrolled courses.



Standard Template Library

Dynamic Vector

The system utilizes `std::vector` as the primary memory store. This allows dynamic resizing during additions and enables fast, contiguous memory iteration during search processes.

Unique Sets

By implementing `std::set` for course enrollments, the underlying Red-Black Tree architecture automatically prevents duplicate courses and maintains alphabetical ordering.

Algorithm Library

The program includes the header to access highly optimized, pre-compiled sorting routines rather than relying on manual, slower sorting implementations.

Record Management Logic

Addition & Validation

The `addStudent` function prevents data corruption by iterating through existing IDs before insertion to block duplicates. It also implements strict mathematical boundary checks for GPA inputs.

$$0.0 \leq \text{GPA} \leq 4.0$$

Retrieval & Deletion

The `searchStudent` function applies linear iteration to locate target IDs rapidly. For removals, `removeStudent` utilizes C++ iterators to safely invoke `vector.erase()` without memory leaks.

Course Enrollment System

Managing Academic Paths

The application completely offloads the complexity of duplicate-checking to the C++ Standard Library.

When the `enrollCourse` function is triggered, it directly invokes the `insert()` method on the Student's course set. If the course already exists, the insertion is naturally ignored. The `showCourses` function then simply traverses this unique set to output the student's curriculum.

Custom Sorting Logic

λ

Lambda Expression

Ranking by Performance

Instead of writing a lengthy custom comparator function, the `sortStudents` feature employs a modern C++ inline Lambda Expression.

This lambda is passed directly into `std::sort`, instructing the algorithm to compare the memory addresses of two Student objects and sort them in descending order based strictly on their `getGpa()` return values.

File I/O & Persistence

saveToFile()

Iterates over the vector upon exit, serializing the state of all objects directly to `students.txt`.

```
:root {
  --main-color: #009bff;
  --bg-color: #fff;
  --text-color: #252525;
  --transition: .2s;
}

.dark-theme {
  --primary-color: #5de4ff;
  --bg-color: #1e1e1e;
  --text-color: #fff;
}

body {
  background-color: var(--bg-color);
  color: var(--text-color);
  transition: background-color var(--transition),
    color var(--transition);
}
```

Format Structure

Generates a parseable string using commas for variables and semicolons exclusively for the course set.

loadFromFile()

Fires on program startup, utilizing `std::stringstream` to rebuild the objects perfectly from text.

Menu Execution Flow

Choice	User Operation	Underlying C++ Implementation
1 & 2	Add / Remove Record	`vector::push_back()` / `vector::erase()`
3 & 4	Search / Display All	Range-based `for` loops scanning the vector
5 & 6	Enroll / View Courses	Locating Student ID, then `set::insert()`
7 & 8	Sort List / Exit App	`std::sort` with Lambda / `ofstream` write

End of Review

Thank you for evaluating the technical implementation.

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Image Sources

https://www.drawio.com/assets/img/blog/dark-mode-default-theme.png

Source: www.drawio.com

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https://wpcdn.web.wsu.edu/wp-vcea/uploads/sites/3427/2026/01/Computer-Science-in-AI-scaled.jpg

Source: school.eecs.wsu.edu

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